Complex Numbers

ANSWER KEY

Section 1: Modulus-Argument Form

1 Find the polar representation of $z = -1 - i\sqrt{3}$.

a. $2(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3})$ ✓

b. $2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

c. $2(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3})$

d. $4(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3})$

$r = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$. Angle is in Quadrant III: $\theta = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$.

2 What is the modulus of the quotient $\frac{1+3i}{2-i}$?

a. $\sqrt{2}$ ✓

b. 2

c. $\sqrt{5}$

d. $\frac{\sqrt{10}}{5}$

$|\frac{1+3i}{2-i}| = \frac{|1+3i|}{|2-i|} = \frac{\sqrt{1^2+3^2}}{\sqrt{2^2+(-1)^2}} = \frac{\sqrt{10}}{\sqrt{5}} = \sqrt{2}$.

3 If $z = 4(\cos 70^\circ + i\sin 70^\circ)$ and $w = 2(\cos 40^\circ + i\sin 40^\circ)$, what is $\frac{z}{w}$?

a. $2(\cos 30^\circ + i\sin 30^\circ)$ ✓

b. $2(\cos 110^\circ + i\sin 110^\circ)$

c. $8(\cos 30^\circ + i\sin 30^\circ)$

d. $2(\cos 2800^\circ + i\sin 2800^\circ)$

Divide moduli: $\frac{4}{2} = 2$. Subtract angles: $70^\circ - 40^\circ = 30^\circ$.

Section 2: De Moivre's Powers

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4 Evaluate $(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}})^{10}$.

a. i ✓

b. $-i$

c. 1

d. -1

$z = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$. Raising to power 10: $\cos(\frac{10\pi}{4}) + i\sin(\frac{10\pi}{4}) = \cos(\frac{5\pi}{2}) + i\sin(\frac{5\pi}{2}) = 0 + i(1) = i$.

5 What is $(\sqrt{3} - i)^6$?

a. -64 ✓

b. 64

c. $-64i$

d. $64i$

$r = 2, \theta = -\frac{\pi}{6}$. Then $z^6 = 2^6[\cos(-\pi) + i\sin(-\pi)] = 64(-1 + 0) = -64$.

6 If $z + \frac{1}{z} = 2\cos\theta$, what is $z^n + \frac{1}{z^n}$?

a. $2\cos(n\theta)$ ✓

b. $2\sin(n\theta)$

c. $\cos(n\theta)$

d. $2^n\cos^n\theta$

Since $z = \cos\theta + i\sin\theta$, $z^n = \cos(n\theta) + i\sin(n\theta)$ and $z^{-n} = \cos(n\theta) - i\sin(n\theta)$. Adding gives $2\cos(n\theta)$.

Section 3: Roots of Complex Numbers

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7 What is the principal square root of $8i$?

a. $2 + 2i$ ✓

b. $2 - 2i$

c. $4i$

d. $2\sqrt{2} + 2\sqrt{2}\,i$

$8i = 8e^{i\pi/2}$. Square root is $\sqrt{8}\,e^{i\pi/4} = 2\sqrt{2}\,(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}) = 2 + 2i$.

8 The four 4th roots of -1 form the vertices of what geometric polygon in the complex plane?

a. **A square** ✓

b. A rectangle

c. A rhombus

d. A regular hexagon

The n-th roots of any non-zero complex number are evenly spaced on a circle with angle separation $\frac{2\pi}{n} = \frac{\pi}{2}$, forming a square.

9 If ω is a non-real cube root of 1, what is the value of $1 + \omega + \omega^2$?

a. 0 ✓

b. 1

c. -1

d. 3

By the properties of roots of unity, the sum of all cube roots of unity is 0.

Section 4: Complex Geometry & Loci

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10 Describe the locus of points in the complex plane satisfying $|z-3| = |z-5i|$.

a. **Perpendicular bisector of segment joining $(3, 0)$ and $(0, 5)$** ✓

b. Circle of radius 3

c. Ellipse with foci at 3 and $5i$

d. Ray starting at 3

The equation states that the distance from z to 3 equals the distance from z to $5i$, which is the perpendicular bisector.

11 What is the Cartesian equation for $|z-2| = 3$?

a. $(x-2)^2 + y^2 = 9$ ✓

b. $x^2 + (y-2)^2 = 9$

c. $(x-2)^2 + y^2 = 3$

d. $x^2 + y^2 = 9$

Let $z = x + iy$: $|(x-2) + iy| = \sqrt{(x-2)^2 + y^2} = 3 \Rightarrow (x-2)^2 + y^2 = 9$.

12 Which inequality describes the interior of the unit disk in the complex plane?

a. $|z| < 1$ ✓

b. $|z| > 1$

c. $\text{Re}(z) < 1$

d. $\text{Im}(z) < 1$

$|z| < 1$ is the set of all complex numbers whose distance to the origin is strictly less than 1.

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